



PAPER ID-311357

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Subject Code: BMI103

Roll No:

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MCA-INT
(SEM I) THEORY EXAMINATION 2024-25
PROBLEM SOLVING USING C

TIME: 3 HRS**M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question	CO	Level
a.	In a 16 bit machine, what is the maximum and minimum integer value can be stored?	2	K2
b.	Differentiate between actual parameter and formal parameter.	3	K2
c.	Determine the value of the following expressions if a=5, b=10, c = -27 i. $a + b * c \% 5$ ii. $a > b \ \&\& \ b > a \ \ a == c$	3	K3
d.	What will be the output of the code? void main() { int a, b, *p; a = 10; p=&a; *p = *p+20; printf("%d", b); printf("%d", a); }	4	K3
e.	Explain the concept of nested structure.	5	K2
f.	Differentiate between dot operator and -> operator to access objects of structure	5	K2
g.	Explain the use of fseek() function.	6	K2

SECTION B**2. Attempt any three of the following:****07 x 3 = 07**

Q no.	Question	CO	Level
a.	Define algorithm and explain the characteristics of Algorithm. Write an algorithm to find the roots of a quadratic equation.	2	K3
b.	How entry-controlled loop is different from exit-controlled loop? Write a program to print the pattern a a p a p p a p p l a p p l e	3	K3
c.	What do you mean by function prototype? How a 2 - dimensional array is passed to a function? Write a function named Add_Matrix which will add two input matrix and displays the result.	4	K3
d.	How a structure is different from an array? Write a C program to store employee details such as Empid, Name, Salary and Age for 50 employees and display the same for the users.	5	K3
e.	Develop a C program to copy the contents of one file to another file. Find the number of word in that file.	6	K3



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SECTION C3. Attempt any one part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	What is the use of flow charts? Explain the symbols used in a flowchart. Draw a flowchart to find the input year is a leap year or not?	2	K3
b.	What do you mean by operator precedence and associativity? Explain left shift and right shift operator with suitable example.	2	K2

4. Attempt any one part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Give the syntax and explain the working of switch-case statement. Write a program to display the month name for an input value 1 to 12 (1 – January, 2 – February etc.) use switch case for implementation	3	K3
b.	Identify the use of modular programming? Write a program by using user define function to check given number is palindrome or not Prototype of function should be like this <code>int is_palindrome (int).</code>	3	K3

5. Attempt any one part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	We want to sort a set of integers. Write a user defined function named BUBBLESORT which will take one array of integers and display the contents in sorted order. (Use Bubble Sort technique)	4	K3
b.	Define a pointer. How do you declare and initialize a pointer? Write a program to add the contents of an integer array using pointer.	4	K3

6. Attempt any one part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Define scope and lifetime of a variable. Explain in detail about all storage classes supported in C language with reference to scope and lifetime.	5	K2
b.	How a structure is different from union? Write a program to store 20 Hotel's details such as Name, City, room rent. For a given city name entered by user, display the hotel details available in that city	5	K3

7. Attempt any one part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Why dynamic memory allocation is better as compared with static memory allocation? Write a program that will allocate memory to store N numbers and find then find the average of those N numbers.	6	K3
b.	Explain the use of the following functions i) fopen() ii) fseek() iii) ftell() iv) feof()	6	K3